Figure V1.16.1

Graded Quality Space irrep decomposition

Linear coordinate sector R^6 (6 DOF) + quadratic stress sector Sym^2(R^6) (21 DOF) = 27 graded DOF under I_h.

symmetric 6x6 stress tensor

s11	s12	s13	s14	s15	s16
	s22	s23	s24	s25	s26
		s33	s34	s35	s36
			s44	s45	s46
				s55	s56
					s66

upper triangle + diagonal = 21 independent components

6 x 7 / 2 = 21

Quadratic sector — Sym^2(R^6) irrep decomposition (21 DOF)

I_h →

A_g	G_g	H_g¹	H_g²	H_g³
1D x2	4D	5D	5D	5D
	Texture coupling	Olfaction & taste	Olfaction & taste	Olfaction & taste

Sym^2(R^6) = 2 A_g + G_g + 3 H_g

2(1) + 4 + 3(5) = 21 (no 3-D irrep)

Linear sector — coordinate space R^6 = E_par (+) E_perp (6 DOF)

T_1 (E_par) 3D	T_2 (E_perp) 3D
Spatial / somatosensory + auditory	Color (trichromatic substrate)

R^6 = T_1 (+) T_2 = 3 + 3 = 6 (the two 3-D irreps)

Irrep	Sector	Dim	Model role in Ch16
A_g	quadratic	1 (x2)	Valence + intensity (two scalar invariants)
G_g	quadratic	4	Texture coupling (olfaction / taste)
H_g¹	quadratic	5	Olfaction & taste / point-defect chemistry
H_g²	quadratic	5	Olfaction & taste / point-defect chemistry
H_g³	quadratic	5	Olfaction & taste / point-defect chemistry
T_1 (E_par)	linear	3	Spatial / somatosensory + auditory
T_2 (E_perp)	linear	3	Color (trichromatic substrate)

Footer discipline: modality labels are Tier 3; the graded I_h decomposition (6 linear + 21 quadratic = 27 DOF) is the structural claim.